Decode & Detect: scoring suspicious XRPL payments

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Code	github.com/manocw/xrpl_hackathon (branch `two-stage-model`); submitted ZIP built from the final commit of our `main` (hash given with the submission)
Final submission	`b1_rank_v7_v12`: site score **74.0** (100 × AP on the test week); rank average of two two-stage models, v7 (site 73.8, grouped CV 62.4) and v12 (site 73.5)

Summary. We score each payment with a two-stage LightGBM model over ~390 behavioural features computed from the whole week of the payment's own split: who funded the sender and how fast, whether the payment is part of a fast multi-hop trail or cycle, whether a sender-receiver pair is a one-off, whether a hub behaves like a scheduled legitimate relay, how the wallet's behaviour deviates from its own history, and the pace of the motif the payment belongs to. A second stage lets each payment borrow the first-stage scores of payments linked to it in time, including two helper models that learn what relays and planted motifs look like. The model goes from the organiser baseline's 29.9 grouped-CV AP to 62.4 and scores 73.8 on the test week; averaging its ranking with a variant built for medium cases gives our final 74.0. Easy motifs are essentially solved; medium and hard cases sit close to an information limit that we measured directly.

1. Data and what drives the problem

Each split is one synthetic week of an XRPL-style ledger: ~40k payments between ~700 wallets in XRP and issued USD/GBP/EUR tokens from two issuers, some converting currency through liquidity providers. Train has 1,250 suspicious payments (3.1%). Wallets, issuer addresses and dates differ between splits, so features must describe *behaviour*, never identity. Payments are valued from the delivered amount at the hourly market rate; the sender's real cost is the first executed transfer leg.

Exploratory analysis (with labels and annotation metadata used for analysis and CV grouping only) produced the findings the whole design rests on:

1. **It is about bad motif instances, not bad wallets.** Labels are pure per planted scenario (~5-28 payments). 169 suspicious scenarios sit among 1,328 legitimate look-alike scenarios of the *same shapes* (fan-out, fan-in, cycle, chain, ...). Suspicious senders send mostly legitimate traffic (~5% suspicious), and 535 of 700 wallets touch something suspicious, so blacklists cannot work.
2. **Speed separates the easy cases.** Easy motifs complete in 2-14 minutes (a 6-hop cycle in 2.4 min); look-alikes take about an hour. Medium motifs are slowed to 1-3 h and overlap the look-alikes.
3. **One-off relationships mark planted edges.** A pair used once in the week is typical of motifs; legitimate commerce repeats 3-7 times at a fixed period and near-constant amount.
4. **Legitimate relays are the largest false-positive source.** 12 cohorts of 100 look-alike relay scenarios run hub-and-spoke rounds all week: 998 of the baseline's top-5% false positives. Relay edges repeat tens of hours apart; half of legitimate relays run "backwards" (hub pays out, is refilled later), while all 12 suspicious relays run forwards and pass on ~95% of each hop.
5. **Raw XRPL fields carry no signal** (fees uniform over 10/12/15 drops, LastLedgerSequence always +20, SendMax always 1.01× cost, in-ledger order random). Only context matters: timing, counterparties, value flow.
6. **Planted motifs are uniform in time** while background traffic is diurnal or in rounds, so quiet periods are enriched for motifs (suspicious and look-alike alike).

2. Methodology

```
parquet (transactions + transfers + hourly rates)
 -> one row per payment: sender cost (first leg), delivered value, USD at the hourly rate
 -> 10 feature families, each computed on one split's unlabelled week only (cached)
 -> stage 1: LightGBM on train labels
 -> stage 1 helpers: LightGBM on auxiliary targets (relay cohort, planted motif); targets only
 -> stage 2: LightGBM re-scores each payment from its linked neighbours' out-of-fold stage-1 scores
 -> final: rank average of v7 and v12 (v7 + medium family + medium specialist), scaled to [0, 1]
```

Design rules: features are relative to the wallet and to the split (ratios, ranks, gaps, recurrence), never addresses, absolute dates or labels. Using the whole unlabelled week of the same split as context is allowed by the rules and is what makes the hard, long-history cases visible at all.

3. Features

Family	What it measures	CV AP alone + starter
starter	Organiser's 12 rules re-implemented: value, sender/receiver activity, forwarding within 1 h, 10-min bursts, pass-through	33.7

funding	Time since the sender was funded and by whom; funder fan-out to wallets that pay on; fan-in/out widths in 5 min-3 h windows; one-off pairs around the payment	58.7
relays	Edge recurrence across rounds, hub in-burst → out-burst matching, forward vs refill direction, value conserved per hop	53.8
fields	Account cadence, pair novelty, amount vs the wallet's own median, network load around the payment	52.4
motifs	Unsupervised reconstruction of motif instances from wallet timelines: size, span, shape, value retention of the payment's group	51.8
hops	Longest time-matched chain through the payment (10 min / 1 h / 6 h), hop delays, value kept per hop, cumulative skim, cash-out hub at the end of the trail	50.8
history	Pair regularity, wallet roles, deviation from the wallet's own week	49.3
amounts	Value relations to adjacent payments (pass-on ratios); static amount tests were null	48.0
cycles	Time-respecting paths back to the sender: cycle length, duration, value drift	45.7
medium (v12 only)	Bursts of rarely used pairs per hub: amount roughness (random order vs the smooth sequences of look-alike fans), gap regularity, one-off "butterflies" (stack width)	-
tempo	Pace of the payment's one-off-pair group (min/median step, regularity, a 12-45 min "slow band") and whether value rises or falls at each hand-off. Medium motifs are slower than look-alikes (median step 17.8 vs 9.5 min); look-alikes lose value at every hop	-

4. Model

- **Stage 1:** LightGBM, binary log-loss, 400 trees, learning rate 0.03, 31 leaves, ≥20 samples per leaf, 80% row and column subsampling, L2 = 1.0, averaged over 3 seeds; 389 features.
- **Stage 2 (score sharing):** for each payment, max / mean / top-2 mean of the stage-1 scores of (a) the same sender's and (b) the same receiver's payments within 10 min, 1 h and 6 h, (c) the receiver's onward payments, (d) the sender's funding payments in those windows, and (e) its one-off-pair motif group (links within 2, 10, 30 min). A second LightGBM with the same settings uses these plus the stage-1 features. It is trained on *out-of-fold* stage-1 scores, and its CV is nested, so no payment ever sees a score produced by a model that saw its label.
- **Auxiliary helper targets (final model):** stage 1 also trains two helper LightGBMs on the same behavioural features: "is this payment part of a relay cohort?" (~20k positives) and "is it part of a planted non-relay scenario, suspicious or legitimate look-alike?" (~2k positives). Their out-of-fold scores get the same neighbour pooling and feed stage 2, which can then tell "a structure" apart from "a bad structure" with far more examples than the 1,250 suspicious labels. **Disclosure:** the targets come from the `cohort_id`, `scenario_id` and `family` annotations in `train/labels.csv`. They are used as training targets only, never as features; at test time nothing from `labels.csv` is needed.
- **Final blend:** v12 adds the medium family and a teammate's *medium specialist* (a stage-1 LightGBM trained on train plus copies of the easy motifs re-timed to medium pace, added at weight 0.3). Alone it scores 73.5; averaging its ranking with v7 (half each) scores 74.0. Rank averaging is reproducible with `src/blend.py`.
- CPU only: ~4 min on a laptop (21 LightGBM fits), peak memory ~1.9 GB. No GPU, external services, pre-trained models or LLMs at runtime.

5. Validation approach

5-fold `StratifiedGroupKFold` on train, grouped by relay cohort > scenario > payment, so whole planted motifs and whole relay cohorts stay in one fold. Random row splits leak sibling payments of the same motif across folds and, with only ~250 positives in a 20% holdout, move the score by several points from split to split. The metric is the site's AP, also reported per difficulty (easy / medium / hard positives against all negatives). Hyperparameters were checked but barely matter (default 60.1, 15 leaves 60.3, 7 leaves 59.2: within the ~0.5 seed noise), so effort went into features and the two-stage design.

6. Results

Model	CV AP	easy	medium	hard	Site score
Organiser starter (logistic regression, its own random split)	24.6				
Starter features, logistic regression, grouped CV	29.9	38.3	1.3	1.1	
Starter features + LightGBM	33.7	44.5	1.4	0.9	
v1: 7 families + LightGBM	59.4	76.7	6.0	1.1	71.0
v2a: 8 families (+ history)	60.1	77.3	6.4	1.1	71.3
v2b: v2a + stage-2 score sharing (nested CV)	61.6	79.6	6.3	1.2	72.2
v2c: v2b + auxiliary relay/motif helper targets	62.3	78.8	8.5	1.2	72.6
v3: v2b + cycles family	62.0	79.9	6.5	1.2	72.3

v6: v3 + auxiliary helper targets (9 families)	62.1	77.9	9.1	1.2	73.0
v7: v6 + tempo family	62.4	78.4	9.2	1.2	73.8
v12: v7 + medium family + 0.3 × medium specialist					73.5
b1: rank average of v7 and v12 (final)					**74.0**

On the test week, v7's top 1% (~401 payments) is 94% precise and holds 30% of all positives: ~50% of easy and ~10% of medium positives (v2c: 0% medium), none of the hard ones. Only ~32% of positives can fit in the top 1% at all. CV moves less than the site (v6 → v7: CV +0.4, site +0.8): 3- vs 10-seed runs of one model share only 369 of 401 top-1% payments, so CV differences of a few tenths are partly seed noise, and we use the site only to confirm changes that CV supports or at least does not reject. Re-runs of v2b and v2c on a second machine reproduced their files exactly. The site score is above our grouped CV, i.e. the behavioural features transfer to a week with new wallets, new issuers and more currency conversions.

7. Interesting findings

- **The relay features are a false-positive killer.** "Does this pair repeat on a schedule?" and "does the hub run forwards?" cut legitimate relays in the top 5% from 998 to 462 and background payments from 206 to 94.
- **Where the remaining errors are** (out-of-fold, v1): 754 of 800 easy positives rank in the top 1,250. The negatives that share that range are mostly legitimate look-alike motifs (329), then relays (109) and background (22). Medium positives sit around rank 2,500, behind ~730 relay and ~660 look-alike payments.
- **The look-alikes are adversarially designed.** A legitimate bipartite look-alike steps every 6-12 s, faster than the easy suspicious bipartite (~40 s); a legitimate multi-stage look-alike outranks an easy suspicious one moving in 12-second steps. Speed alone cannot separate them; only combinations of context can.
- **Medium is at the information limit.** Given the *true* scenario groupings and 22 whole-scenario statistics (analysis only), a classifier separates medium suspicious from look-alike scenarios at AUC 0.48-0.57, i.e. chance, versus 0.87-0.91 for easy. The only weak signal: medium motifs are *slower* than their look-alikes (median step 15-18 vs ~9 min, AUC 0.70) and keep more value per hop (0.91-0.96 vs 0.83-0.88).
- **Hard relays likewise.** With the true relay scenarios, the only property separating the 12 suspicious relays from 1,188 legitimate ones is direction (all forwards vs half refills, AUC 0.75); size, rounds, counterparties, retention, amounts and conversions are all at AUC ~0.5.
- **Telling the model the pace pays off on the test week.** A model trained mostly on easy (fast) cases learns "fast = suspicious" and ranks slow medium motifs below look-alikes. Stating each motif group's pace and hop-value pattern explicitly (tempo) lifted the site score from 73.0 to 73.8 and brought medium payments back into the top 1%.
- **Learning "structure" separately from "bad" helps medium cases.** The relay/motif helper targets lift medium AP from 6.3 to 8.5 in CV, the only change that moved medium at all.
- **Score sharing helps where motifs are clear.** Stage 2 adds +1.5 to +2.3 CV AP, nearly all of it on easy cases: once part of a fast motif is found, its siblings follow.

8. Tested and rejected ideas

Each of these was measured with the grouped CV or on the leaderboard and dropped because it did not help (differences of ≤0.5 AP are within seed noise).

Idea	Result	Why we think it failed
Currency-conversion features (legs, spreads, cross-issuer swaps)	Excluded by design	1/2/3-leg mix identical for suspicious and legitimate (59/31/10%); conversion share shifts from 41% (train) to 50% (test), so these features would only add drift.
Static amount tests: round amounts, just under 100/500/1k/3k/5k/10k thresholds, repeated identical amounts	Null	No excess mass under any threshold; amounts top out ~4.6k USD. Only relations to adjacent payments (pass-on ratios) carry signal.
Hour-of-day profiles	Null	Motifs are uniform in time; the useful version is "network load around the payment".
Mule-wallet reuse across scenarios	Null	Wallets in medium/hard scenarios are no more linked to other suspicious scenarios than look-alike wallets (6.9% vs 6.6%).
A third helper target "medium suspicious" (225 positives)	CV 62.3 vs 62.1; on v7: site 73.3 vs 73.8	Too few medium positives for a separate scorer; tempo carries the pace signal better.
Medium specialist or medium family added to v7 on their own	Site 73.2 / 73.3 vs 73.8	Each lifts slow look-alikes as well; only as one half of a rank average do they help (74.0).
Adding v6 to the final rank average	Site 73.8 vs 74.0	v6 is already inside v7.
Self-training stage 2 on the test week (top 1.5% / bottom 50%, 2 rounds)	Site 72.2 vs v3 72.3	Reinforces the easy cases the model already ranks well.

Graph neural network on the transaction graph (Colab)	AP 49.2 vs LightGBM 63.6 on a quarter week	Trees on engineered features win at this data size; +0.7 only as a 10% blend.
Motif context: funding before, cash-out after, common funder or destination	AUC 0.45-0.56 with true groupings	The generator plants motifs with no surrounding laundering structure.
Reconstructing medium motifs by shared wallets and scoring the group	No gain	Grouping bleeds into background traffic: one-off-pair groups are only 58% pure at a 30-min link, 37% at 1 h.
Hyperparameter tuning (leaves, min samples per leaf)	59.2-60.3 vs 60.1	Within seed noise; the limit is information, not capacity.
Weighting suspicious payments ×2 (`scale_pos_weight` 2) in both stages	CV 60.5 vs 61.6	AP depends only on ranking; extra weight made stage 2 overfit the positives (medium AP 5.8 vs 6.3).
Teammate's flow-episode model (site 66.8) in one model with ours, or rank-blended in CV	CV 60.3 / 60.4 vs 60.1	Both feature designs capture the same signal.
Rank blend on test: 0.85 v2b + 0.15 flow-episode model; 0.70 / 0.30	Site 71.7; 71.2 (v2b 72.2)	Rankings agree at the top (276/401 shared in the top 1%, Spearman 0.59); the disagreement is in the bulk, where the other model is weaker. More weight, lower score.
Self-training on the test week: top 300 test payments as pseudo-positives, bottom 60% as pseudo-negatives, 4 time blocks cross-fitted, weight 0.5	Site 71.5; blended 50/50 with v2b 72.1	Pseudo-labels repeat what the model already ranks well (easy cases); they add nothing for medium/hard, which are at their information limit.

9. Limitations

- Medium cases remain hard (AP ~9) and hard cases near random (AP ~1): slowed-down motifs look like their legitimate look-alikes, and we showed this holds even with perfect grouping.
- 25 "ambiguous" hard payments sit inside regular scheduled series with the same amount and period as their legitimate siblings: unlearnable by construction.
- Distribution drift: test has more tiny payments (27% under $20 vs 22%) and more conversions. We prefer ratio and retention features over absolute amounts for that reason.
- Whole-week per-wallet aggregates could memorise wallets inside train and so inflate CV; wallets are disjoint across splits, so this cannot leak into test (and the site score is above CV).
- The data is synthetic: the patterns are the simulator's. Transfer to the real XRPL is untested.
- Future work: a temporal edge-level graph neural network with GPU time. Our quick GNN trailed the trees; stage 2 is already one hand-built round of message passing.

10. Reproducibility

Python 3.14, CPU only. Put the organiser data under `data/train/` and `data/test/`, then:

```
python -m venv venv && ./venv/bin/pip install -r requirements.txt
./venv/bin/python -m src.predict --features starter funding relays fields motifs hops amounts history \
    cycles tempo --seeds 3 --stack --aux --name v7_10fam_tempo_aux
./venv/bin/python -m src.predict --features starter funding relays fields motifs hops amounts history
cycles tempo medium --seeds 3 --stack --aux --name v11_11fam_tempo_medium_aux
./venv/bin/python -m src.specialist --base v11_11fam_tempo_medium_aux --weight 0.3 --name v12_v11_plus_spec
./venv/bin/python -m src.blend submissions/v7_10fam_tempo_aux.csv submissions/v12_v11_plus_spec.csv     --
weights 0.5 0.5 --name b1_rank_v7_v12           # -> submissions/b1_rank_v7_v12.csv, ~20 min
```

Add `--cv` for the nested grouped CV report. Libraries: pandas, numpy, pyarrow, scikit-learn, LightGBM, networkx, scipy. Labels and annotation metadata are used only as training targets (main and helper models) and for CV grouping, never as features; no external data. The code was written with AI assistance (Claude Code) under the team's direction and review; no LLM is part of the model. Licence: MIT.